

Digital Logic Design Report

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Course: Principles of Computer Systems Laboratory

1 Binary to Gray and Gray to Binary Code Converter

In this project, a converter was implemented to transform a 4-bit binary number into its corresponding Gray code and vice versa. The conversion formulas used are as follows:

- **Binary to Gray:** The most significant bit (MSB) remains unchanged. Each subsequent bit is obtained by XORing the current binary bit with the previous one.
- **Gray to Binary:** The MSB remains unchanged. Each subsequent binary bit is found by XORing the previous binary bit with the current Gray code bit.

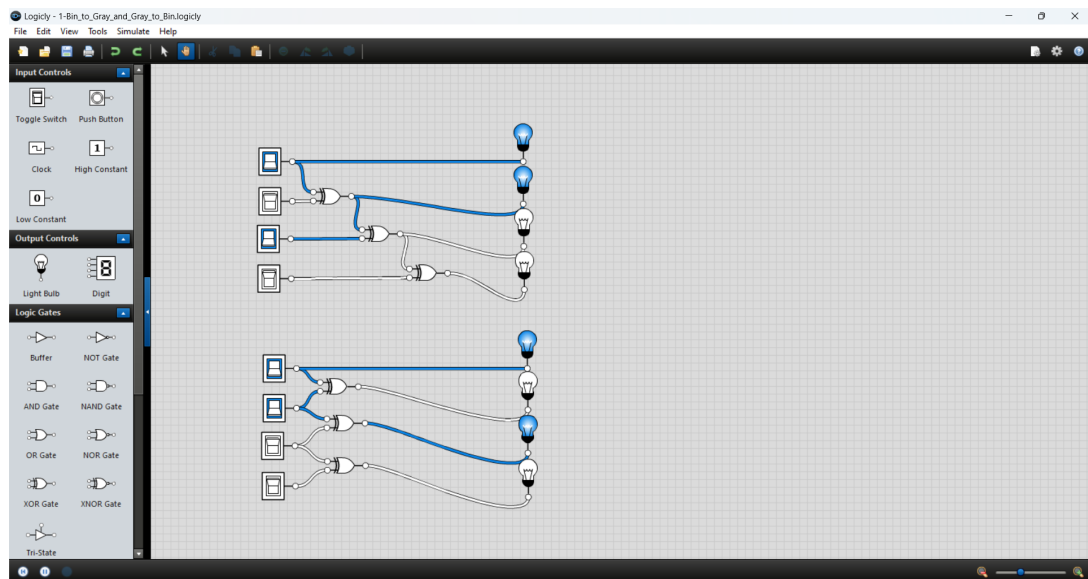


Figure 1: Binary to Gray and Gray to Binary Conversion Circuit

The circuit was tested successfully with two different 4-bit values to verify accurate conversion in both directions.

2 Full Adder Design

The full adder circuit was designed and implemented using basic logic gates. The full adder adds two input bits and a carry-in bit, producing a sum and a carry-out bit.

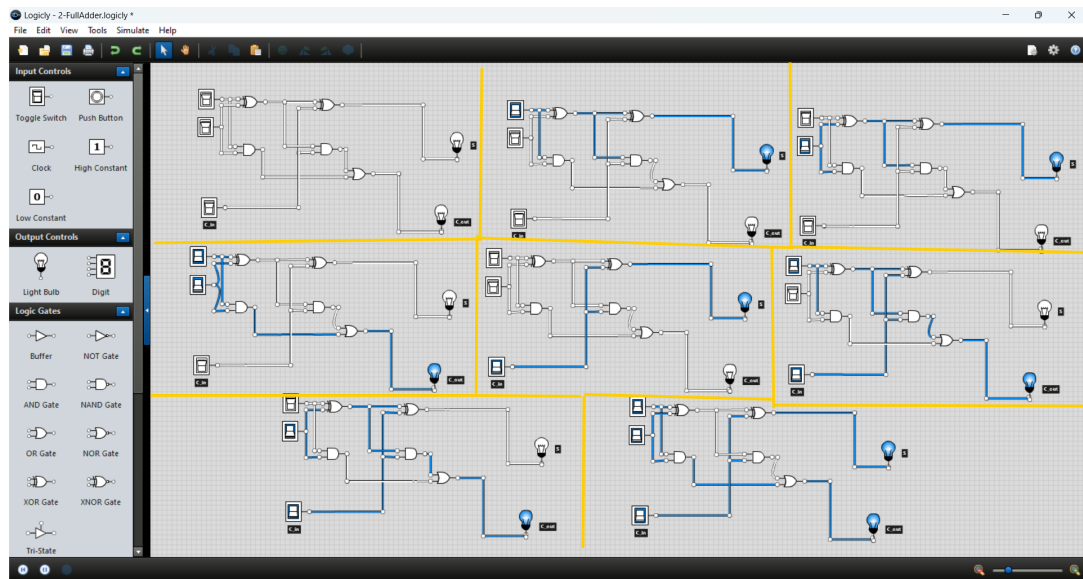


Figure 2: Full Adder Circuit Designs

The circuit was tested across all 8 possible combinations of 3 input bits (A, B, and Carry-in), verifying the correct sum and carry-out results.

3 3-bit Binary Adder Using Full Adders

A 3-bit binary adder was constructed by chaining three full adders in series. Each full adder handles a corresponding bit from the input binary numbers.

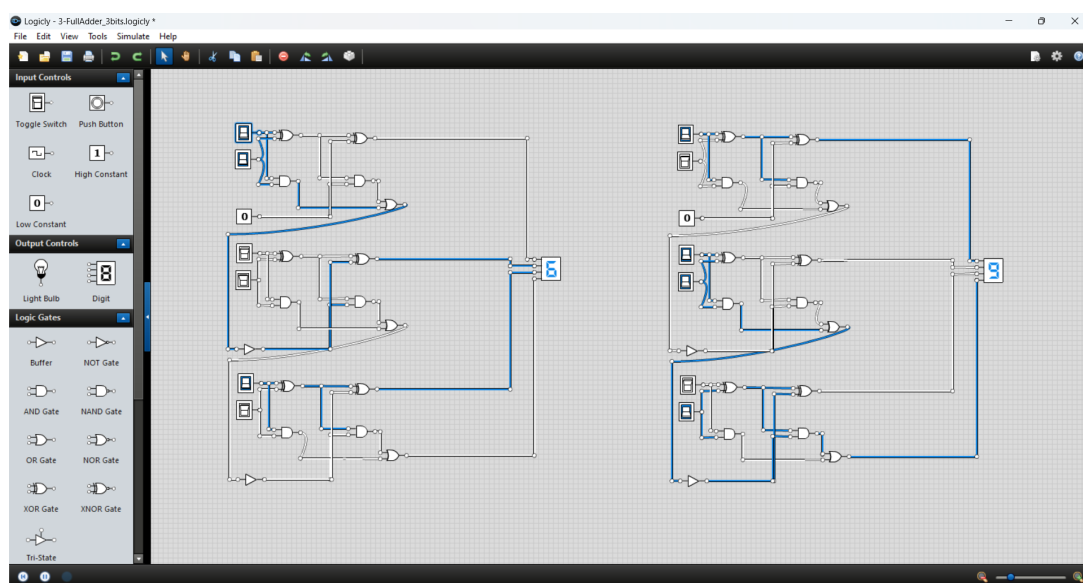


Figure 3: 3-bit Full Adder Circuit

This design was tested with different 3-bit binary numbers to validate the overall addition result and the final carry-out.

4 Conclusion

In this laboratory project, we explored various fundamental digital logic circuits:

- Code converters (Binary $\leftrightarrow$ Gray)
- Full adder circuits
- Multi-bit binary adders using chained full adders

Each design was implemented and simulated successfully using LogiCly software. These circuits form the foundation of digital arithmetic units in computer systems.